Introduction to Augmented Reality EN.601.654 HW 3

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1. Conceptual:

a. An observable is an event stream that tracks listener and notifies when event occurs. An observer is an individual callback function attached to that observable. They work together when an observable executes notification due to an event, which iterates over all registered observers and run their callback functions.

b. I would choose pancake lenses, they maximizes clarity in all areas and eliminate chromatic aberration or artifacts typical of Fresnel lenses. The disadvantage would be the low optical light transmission rate due to polarizing beam splitters and higher cost of manufacturing. The headsets would need a really bright screen, which requires more thermal dissipation as well as bigger batteries.

c. I would choose WebXR, only weblinks are needed without any installation. Advantage: immediately accessible from all kinds of devices with a browser. Disadvantage: Lower rendering performance and memory headroom compared to offline counterparts.

d. The shader is rendering the surface with sphere mapping to simulate reflections.

- View vectors are firstly computed, it transforms the fragment position and normal into viewer's coordinate and normalize to obtain view direction *e* and normal *n*;
- Then the view-space reflection vector *r* = reflect(*e,n)*;
- Then *m* projects the 2D reflection direction *r* onto the 2D viewing disk, with 0.5 units of shift to validate UV coordinates [0.0, 1.0];
- Finally, the resulting UV coordinates vN are used for look up the lighting and reflection color from the refSampler.

e. Clip-space coordinates are four-dimensional homogeneous coordinates resulting from the projection transformation in a graphics pipeline, used by the GPU to efficiently clip primitives outside the viewing frustum.

f. It is more memory efficient for GPU rendering. As most of the 3D polygons are not see-through, there is no point using 3D textures if internal volumes are not represented. Through UV unwrapping, 3D shapes can be mapped onto planer 2D surface efficiently and rendered faster than that of a 3D texture.

g. Yes we can shear an object similar to that in optical lenses by applying a shear matrix:

```
mat4 shearY = mat4(
    1.0, 0.0, 0.0, 0.0,
    k,   1.0, 0.0, 0.0,
    0.0, 0.0, 1.0, 0.0,
    0.0, 0.0, 0.0, 1.0
);
vec4 shearedPosition = shearY * vec4(position, 1.0);
gl_Position = worldViewProjection * shearedPosition;
```

Q3. a $u = \frac{x_{ndc}+1}{2} W$, $v = \frac{y_{ndc}+1}{2} H \Rightarrow x_{ndc} = \frac{2u-W}{W}$, $y_{ndc} = \frac{2v-H}{H}$ ①

for point $P_c = \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}$ w/ $z < 0$

$u = f_x \frac{X_c}{-Z_c} + O_x$, $v = f_y \frac{Y_c}{-Z_c} + O_y$ ②

Sub ② into ①

$$x_{ndc} = \frac{\frac{2f_x}{W} X_c + \frac{2O_x - W}{W}(-Z_c)}{-Z_c} \quad , \quad y_{ndc} = \frac{\frac{2f_y}{W} Y_c + \frac{2O_y - H}{H}(-Z_c)}{-Z_c}$$

$$M_{proj} = \begin{bmatrix} \frac{2f_x}{W} & 0 & \frac{2O_x - W}{W} & 0 \\ 0 & \frac{2f_y}{W} & \frac{2O_y - H}{H} & 0 \\ 0 & 0 & -\frac{f+n}{f-n} & -\frac{2fn}{f-n} \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

b. $M_{proj} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & \frac{8}{3} & \frac{1}{3} & 0 \\ 0 & 0 & -\frac{101}{99} & -\frac{40}{99} \\ 0 & 0 & -1 & 0 \end{bmatrix}$

c. for $P^c = (0.3, -0.15, -6.0, 1)^T$

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & \frac{8}{3} & \frac{1}{3} & 0 \\ 0 & 0 & -\frac{101}{99} & -\frac{40}{99} \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0.3 \\ -0.15 \\ -6 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.6 \\ -2.4 \\ \frac{566}{99} \\ 6 \end{bmatrix}$$

$$P_{ndc} = \begin{bmatrix} x_{ndc} \\ y_{ndc} \\ z_{ndc} \end{bmatrix} = \frac{1}{W_{clip}} \begin{bmatrix} X_{clip} \\ Y_{clip} \\ Z_{clip} \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 0.6 \\ -2.4 \\ \frac{566}{99} \end{bmatrix} = \begin{bmatrix} 0.1 \\ -0.4 \\ 0.95 \end{bmatrix}$$

d. $u = 0.5(x_{ndc}+1) W = 0.5(0.1+1) \times 800 = 440$ px

$v = 0.5(y_{ndc}+1) H = 0.5(-0.4+1) \times 600 = 180$ px

4a Figure 1:



4b Figure 1:
Changes: uMatAmbient uMatDiffuse uMatSpecular uMatShininess



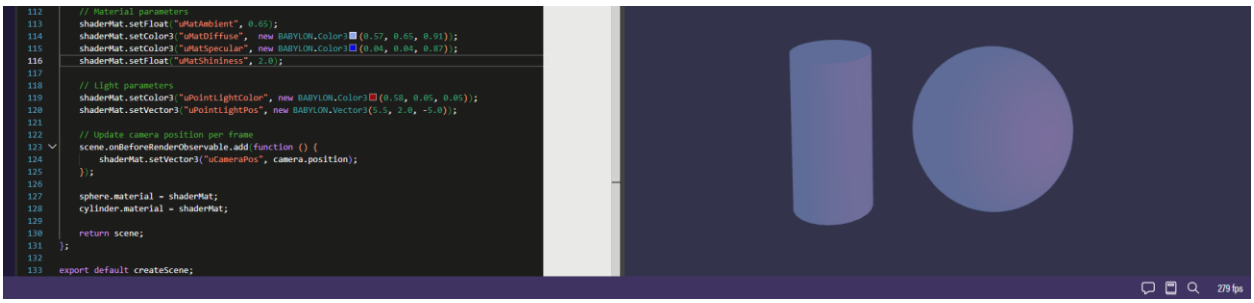
Figure 2:

Changes: uMatDiffuse uMatSpecular uMatShininess uPointLightColor



Figure 3:

Changes: uMatSpecular uMatShininess uPointLightColor uPointLightPos



4c Figure 1: added second light

